DE LA SALLE UNIVERSITY - DASMARIÑAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
COMPUTER SCIENCE DEPARTMENT

COURSE SYLLABUS

Second Semester, SY 2024-2025

COURSE CODE	**:**	S-CSPC327
COURSE TITLE	**:**	Automata Theory and Formal Languages
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	Three units
MODALITY	**:**	Traditional blended learning model
PREREQUISITE	**:**	S-CSPC315 - Algorithm and Complexity
PROFESSOR	:	MS. Jennylinde R. Manaois
CONSULTATION HOURS	:	M / 12:30 - 1:00 / online, T / 8:00 - 12:00 / online, H / 1:00 - 2:30 / COS100c, F / 8:00-12:00 / online
CONTACT DETAILS	:	jrmanaois@dlsud.edu.ph

VISION:

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION:

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

Analyzes formal models in computer science, including regular expressions, finite automata, contextfree grammars, pushdown automata and Turing machine.

COURSE LEARNING OUTCOMES:

By the end of this course, students are expected to:

CLO1. acquire a fundamental understanding of the core concepts in automata theory and formal languages;
CLO2. analyze and construct regular expressions to define regular languages, solve string-matching problems, and effectively describe their corresponding languages in natural language;
CLO3. classify the different computing machines by discussing the different formal models;

CLO4. design grammars and recognizers for different formal languages;
CLO5. explain the principles of pushdown automata (PDAs) by determining the languages they recognize, and designing PDAs for given languages to model computational processes;
CLO6: evaluate Turing machines by identifying their components, their operations, and the languages they recognize; and
CLO6. demonstrate 21st century skills in all learning activities;

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Acquire a fundamental understanding of the core concepts in automata theory and formal languages.	• TLO1. Demonstrate understanding of automata theory and apply its concepts to solve computational problems, including recognition of regular and context-free languages.
CLO2. Analyze and construct regular expressions to define regular languages, solve string-matching problems, and effectively describe their corresponding languages in natural language.	• TLO2. Understand the concepts of regular expressions and apply them to define regular languages and solve string-matching problems. • TLO3. Describe in plain English the language of regular expressions with straightforward descriptions. • TLO4. Write regular expressions of given regular languages.
CLO3. Classify the different computing machines by discussing the different formal models.	• TLO 5. Recognize and describe the components of a Deterministic Finite Automaton (DFA), including states, alphabet, transitions, start state, and accepting states. • TLO 6. Identify the languages of given DFAs. • TLO 7. Construct DFAs based from the given language.

Module No and Learning Outcomes	**Date**	**Teaching-Learning Activities / Assessment Strategy**	**Technology Enabler**	**Onsite**	**Asynchronous**	**Allotted Hours**
Module 1 TLO1 CLO1	Week 1 (Jan.20-24)	• Gospel reading, Reflection and Prayer • Presentation of course introduction • Module 1: Introduction to Computer Theory and the Central Concepts of Automata Theory • Enabling assessment: Assignment (15 pts.)	PowerPoint, Multimedia Devices (Laptop, Projector)	/		3.0

Module 1 TLO1 CLO1	Week 2 (Jan.27-Jan.31)	• Gospel reading, Reflection and Prayer • Module 1: Introduction to Computer Theory and the Central Concepts of Automata Theory • Enabling assessment: Short Quiz on Introduction to Computer Theory and the Central Concepts of Automata Theory (15 pts.)	Schoolbook, Powerpoint		/	3.0
Module 2 TLO 2,3,4 CLO 2	Week 3 (Feb.3-7)	• Gospel reading, Reflection and Prayer • Review of Introduction to Computer Theory and the Central Concepts of Automata Theory • Module 2: Regular Expression • Enabling assessment: Short Quiz on Regular Expression (15 pts.)	PowerPoint, Multimedia Devices (Laptop, Projector)	/		3.0
Module 2 TLO 2,3,4 CLO 2	Week 4 (Feb.10-14)	• Gospel reading, Reflection and Prayer • Review of Regular Expression • Formative assessment: Exercises on Regular Expression (15 pts.)	Schoolbook, Powerpoint		/	3.0
Module 1,2 TLO 1,2,3,4 CLO 1,2	Week 5 (Feb.17-21)	• Gospel reading, Reflection and Prayer • Enabling assessment: Long Quiz on Module 1,2 (40 pts.)	Schoolbook	/		3.0
Module 3 TLO 5,6,7 CLO 3	Week 6 (Feb.24-Feb.28)	• Gospel reading, Reflection and Prayer • Module 3: Finite Automata • Formative assessment: Exercises on Finite Automata(15 pts.)	PowerPoint, Multimedia Devices (Laptop, Projector)		/	3.0
Module 3 TLO 5,6,7 CLO 3	Week 7 (Mar.3-7)	• Gospel reading, Reflection and Prayer • Review of Finite Automata • Enabling assessment: Short Quiz on Finite Automata(15 pts.)	Schoolbook, Powerpoint	/		3.0
Module 3 TLO 5,6,7 CLO 3	Week 8 (Mar.10-14)	• Gospel reading, Reflection and Prayer • Review of Finite Automata • Preparation and Presenation of the 1st part of the Project	PowerPoint, Multimedia Devices (Laptop, Projector)		/	3.0
Module 1,2,3 TLO 1,2,3,4,5,6,7 CLO 1,2,3	Week 9 (Mar.17-21)	**MIDTERM EXAMINATION** Written Examination covering Modules 1-3		/		3.0
			TOTAL	***15.0***	***12.0***	***27.0***

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes

CLO4. Design grammars and recognizers for different formal languages	• TLO8. Understand the concepts of context-free grammar. • TLO9. Describe in plain English the languages of CFGs with straightforward descriptions. • TLO10. Construct CFGs for context-free languages. • TLO11. Argue why a language description is or is not the language of a CFG.
CLO5. Explain the principles of pushdown automata (PDAs) by identifying their components, determining the languages they recognize, designing PDAs for given languages, and converting between PDAs and context-free grammars (CFGs) to model computational processes.	• TLO12. Recognize the components of a pushdown automaton particularly its use of an infinite stack. • TLO13. Identify the languages accepted by given PDAs. • TLO14. Design PDAs of given languages. • TLO15. Convert between PDAs and CFGs.
CLO6. Evaluate Turing machines by identifying their components, their operations, and the languages they recognize.	• TLO16. Identify the components of Turing Machines. • TLO17. Understand how Turing machines operate. • TLO18. Determine the languages accepted by given Turing machines. • TLO19. Construct Turing machines that accept regular languages, context-free languages and some non-context free languages.
CLO7. Demonstrate 21st century skills in all learning activities.	• TLO20. Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments).

Module No and Learning Outcomes	**Date**	**Teaching-Learning Activities / Assessment Strategy**	**Technology Enabler**	**Onsite**	**Asynchronous**	**Allotted Hours**
Module 4 TLO8,9,10, 11 CLO 4	Week 10 (Mar.24-28)	• Gospel reading, Reflection and Prayer • Module 4: Contex Free-Grammar • Sustainability Week	PowerPoint, Multimedia Devices	/		3.0
Module 4 TLO8,9,10, 11 CLO 4	Week 11 (Mar.31-Apr.4)	• Gospel reading, Reflection and Prayer • Module 4: Contex Free-Grammar • Enabling assessment : Short Quiz on Context Free-Grammar (15 pts.)	Schoolbook		/	3.0
Module 5 TLO12,13, 14,15 CLO 5	Week 12 (Apr.7-11)	• Gospel reading, Reflection and Prayer • Module 5: Pushdown Automata • Enabling assessment : Short Quiz on Pushdown Automata (15 pts.)	PowerPoint, Multimedia Devices (Laptop, Projector)	/		3.0

	Week 13 (Apr.14-18)	Independent Learning Week			/	3.0
Module 5 TLO12,13, 14,15 CLO 5	Week 14 (Apr.21-25)	• Gospel reading, Reflection and Prayer • Module 5: Pushdown Automata • Research Week	Schoolbook, Powerpoint		/	3.0
Module 4,5 TLO8,9,10, 11,12,13,14 ,15 CLO 4,5	Week 15 (Apr.28-May2)	• Gospel reading, Reflection and Prayer • Enabling Assessment: Long Quiz on Module 4,5 (40 pts.)	PowerPoint, Multimedia Devices (Laptop, Projector)	/		3.0
Module 6 TLO16,17, 18,19 CLO 6	Week 16 (May5-9)	• Gospel reading, Reflection and Prayer • Module 6: Turing Machine	Schoolbook	/		3.0
Module 6 TLO16,17, 18,19 CLO 6	Week 17 (May12-16)	• Gospel reading, Reflection and Prayer • Module 6: Turing Machine • Enabling assessment: Short Quiz on Turing Machine (15 pts.)	PowerPoint, Multimedia Devices (Laptop, Projector)		/	3.0
Module 4,5,6 TLO8,9,10, 11,12,13,14 ,15,16,17,1 8,19 CLO 4,5,6	Week 18 (May19-23)	**FINALS EXAMINATION** Presentation of Project		/		1.5 1.5
			TOTAL	***15.0***	***12.0***	***27.0***

FINAL PRODUCT/PERFORMANCE:

At the end of the course, the students are expected to develop a simple compiler that will check the syntax of a certain string if it is a valid string or not in a particular language.

RUBRIC FOR EVALUATING THE FINAL PRODUCT:

Criteria	Percentage
Correctness (appropriate algorithm)	60%
Clarity of design & presentation	20%
Finiteness & absence of ambiguity	20%

GRADING SYSTEM

Midterm

Enabling Assessments - 50%
Attendance/Class Participation - 20%
Summative Assessment - 30%

100%

Finals

Enabling Assessments	- 50%
Attendance/Class Participation	- 20%
Summative Assessments	- 30%
	100%

Overall Final Grade = $\frac{\text{Midterm + Final}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    c. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy

a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for student's grade.
b. The onsite meeting will only be held at the schedule indicated in the registration form.
c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones cannot also be used as calculator during examination.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES:

Online References

E-provider	References
Online	Tutorialspoint. 2025. Automata theory tutorial [Internet]. Available from: https://www.tutorialspoint.com/automata_theory/index.htm
Online	YouTube. 2025. Automata: Alphabet, String and Language (Introduction) [Internet]. Available from: https://www.youtube.com/watch?v=BrDowMQljl0

Online	YouTube. Introduction to Theory of Computation [Internet]. 2025 Available from: https://www.youtube.com/channel/UCQYMhOMi_Cdj1CEAU-fv80A

Journals

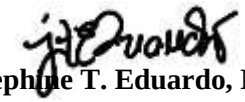
Provider	References
Journal	Robson EW, Ruggerio S, Erickson J. FSM Builder: a tool for writing autograded finite automata questions. *[Journal Name]* [Internet]. 2024 Jul 3 [cited YYYY Mon DD];269–275. Available from: https://dl.acm.org/doi/pdf/10.1145/3649217.3653599
Journal	Aiswarya C, Arvind V, Saurabh S. Theory research in India. *Communications of the ACM* [Internet]. 2022;65(11):88–93. Available from: https://doi.org/10.1145/3550496
Journal	Broy M, Brucker A, Fantechi A, Gleirscher M, Havelund K, Kuppe MA, Mendes A, Platzer A, Ringert J, Sullivan A. Does every computer scientist need to know formal methods? *Formal Aspects of Computing* [Internet]. 2024 Jun 10;[journal volume if available]:[page numbers if available]. Available from: https://doi.org/10.1145/3670795
Journal	D'Antoni L, Veanes M. Automata modulo theories. *Communications of the ACM* [Internet]. 2021;64(5):86–95. Available from: https://doi.org/10.1145/3419404

On-Site References

Provider	References
AEA Lib QA 267.5.S4 .M364 2011	Martin J.. Introduction to languages and the theory of computation. Singapore: McGraw-Hill; 2011.
AEA Lib QA 267 .H77 2001	Hopcroft M, Ullman. Introduction to automata theory, languages, and computation. 2nd ed. Singapore: Addison-Wesley Publishing Company, Inc.; 2001.

AEA Lib QA 76.5 .C66 1997	Cohen D. Introduction to computer theory. 2nd ed. Singapore: Wiley & Sons, Inc.; 1997.
AEA Lib QA 267 .L586 1998	Lewis P. Elements of the theory of computation. 2nd ed. Singapore: Prentice Hall.; 1998.
AEA Lib QA 37.3 .D63 2019	Rosen K. Discrete mathematics and its applications. Singapore: McGraw-Hill.; 2019.

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